

Week 1 — Optimization Is Everywhere, LP is a Convex Problem

OL7014 Integer Programming • Instructor: Prof. H. Park • Aug 24, 2026

1. General optimization problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & x \in X \end{array}$$

1. what do we control? **decision variables** x
2. what do we want? **objective function** f
3. what limits us? constraints $\rightarrow$ **feasible set** X

$$X = \{x \in \mathbb{R}^n : g_i(x) \leq 0 \ \forall i \in [m]\}$$

in plain English: x belongs to the n -dimensional real space $\mathbb{R}^n$, **and** it satisfies all m constraints

$$g_1(x) \leq 0, \quad g_2(x) \leq 0, \quad \dots, \quad g_m(x) \leq 0$$

read the symbols: $\in$ = “is in”, $\forall$ = “for all”, $[m] := \{1, \dots, m\}$, s.t. = “subject to”

— so “ $\forall i \in [m]$ ” packs that whole list of m constraints into one stroke

$$h(x) = 0 \iff h \leq 0 \text{ and } -h \leq 0$$

takeaway how we define $f(x)$ and X determines the program class. some problems are easy to solve — but some are not.

2. One problem, three languages

	OR	ML	control
decision	x	θ	u
uncertain data	ξ	[x = input, y = output] state x perturbed by noise w	

KEY optimization is not a topic inside OR — it is the shared language of OR, ML, control.

watch Brunton, *Optimization: A Bootcamp for Machine Learning, Inverse Problems, and Control* — 22 min, supplementary.
https://www.youtube.com/watch?v=1PBPbGmw1_4

3. Linear programs — and why they are easy

$$\begin{array}{ll} \max_x & c^\top x \\ \text{s.t.} & Ax \leq b \end{array}$$

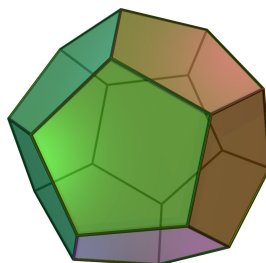
$$c \in \mathbb{R}^n, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m$$

$$\text{row } i: \quad g_i(x) = a_i^\top x - b_i \leq 0 \iff a_i^\top x \leq b_i$$

$$\text{stack rows} \Rightarrow A \quad \text{so} \quad a_i^\top x \leq b_i \ \forall i \iff Ax \leq b$$

in many cases variables are nonnegative ($x \geq 0$) — encoded in $Ax \leq b$ by the rows $-x_j \leq 0$

$P = \{x : Ax \leq b\}$ is called a **polyhedron**



Polyhedron in $\mathbb{R}^3$. (image: Wikimedia Commons)

Convexity

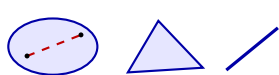
- **Convex set:** $S \subseteq \mathbb{R}^n$ is convex if $\lambda x + (1 - \lambda)y \in S$ for all $x, y \in S$ and all $\lambda \in [0, 1]$
- **Convex function:** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if
$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \text{ for all } x, y \in \mathbb{R}^n \text{ and all } \lambda \in [0, 1]$$
strictly convex if strict for $x \neq y, \lambda \in (0, 1)$
- **Convex problem:** $\min\{f(x) : x \in S\}$ with S convex and f convex

Prop. in a convex problem, every local min is a global min.

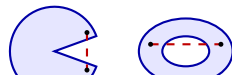
pf sketch: if $f(y) < f(x^*)$, then $f(\lambda y + (1 - \lambda)x^*) \leq \lambda f(y) + (1 - \lambda)f(x^*) < f(x^*)$
for all $\lambda \in (0, 1]$ — so better points sit arbitrarily close to x^* .

Examples

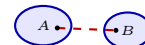
convex	not convex
least squares $\ Ax - b\ _2^2$ over unconstrained $\mathcal{X}$ — one bowl	neural-net training loss — many dips
linear $c^\top x$ over P — our LP	$\in \mathbb{Z}$: feasible set turns discrete ! — next lecture



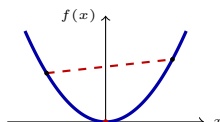
convex sets
every segment stays inside
(a segment is convex too)



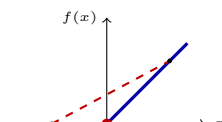
nonconvex sets
some segment leaves the set
(a notch or a hole breaks convexity)



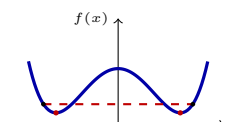
A convex, B convex,
but $A \cup B$ is **not** convex
(union breaks it; intersection never does)



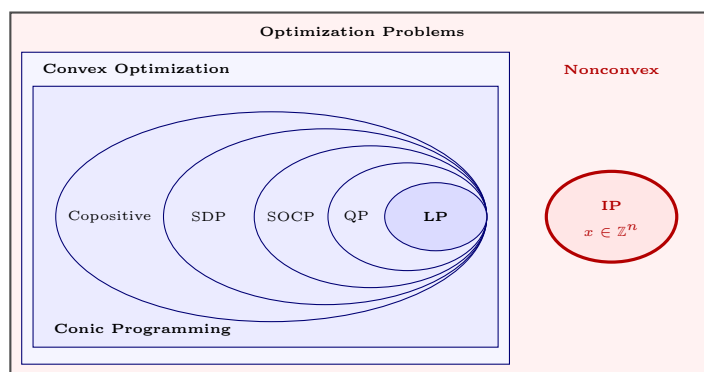
convex — $f(x) = x^2$
chord sits *above* the graph
one bottom, and it is *the* bottom



convex — ReLU $\max(0, x)$
chord still above, but there is a *kink*
convex $\neq$ smooth



not convex
chord dips *below* the graph
two bottoms — which did you find?



$LP \subset QP \subset SOCP \subset SDP \subset \text{Copositive} \subset \text{Conic} \subset \text{Convex}$, but **IP** $\not\subset$ Convex
(QP = quadratic, SOCP = second-order cone, SDP = semidefinite)
local = global holds everywhere inside the convex box — **IP sits outside it: next lecture**

takeaway an LP is a convex problem: P convex (intersection of half-spaces) + $c^\top x$ linear, hence a convex function.

4. First LP problem

x = crates of ammo (dense), y = crates of rations (bulky)

$$\begin{array}{ll}\max & 4x + 5y \\ \text{s.t.} & 30x + 90y \leq 300 \quad (\text{volume, L}) \\ & 300x + 100y \leq 1000 \quad (\text{weight, kg}) \\ & x, y \geq 0\end{array}$$

divide by 30, by 100 $\Rightarrow$

$$\begin{array}{ll}\max & 4x + 5y \\ \text{s.t.} & x + 3y \leq 10 \\ & 3x + y \leq 10 \\ & x, y \geq 0\end{array}$$

now put it in standard vector form:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c = \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 10 \\ 0 \\ 0 \end{pmatrix} \quad (\text{last two rows are } x, y \geq 0)$$

$$\begin{array}{ll}\max & c^\top \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \leq b\end{array} \quad \text{--- Section 3, now with numbers in it}$$

vertices: $(0, 0)$, $(\frac{10}{3}, 0)$, $(2.5, 2.5)$, $(0, \frac{10}{3})$

slide $4x + 5y = z$ up until it leaves $\Rightarrow \boxed{(x^*, y^*) = (2.5, 2.5), \quad z_{\text{LP}} = 22.5}$

KEY 1. an optimum sits at a vertex.

2. fractional answers are normal — nothing forces an LP onto whole numbers.

demo `demo1a_lp_graphical.py` — Play animation slides the line; every vertex labeled with $c^\top v$.

next lecture one new line, $x, y \in \mathbb{Z}$ — and what it does to everything above.